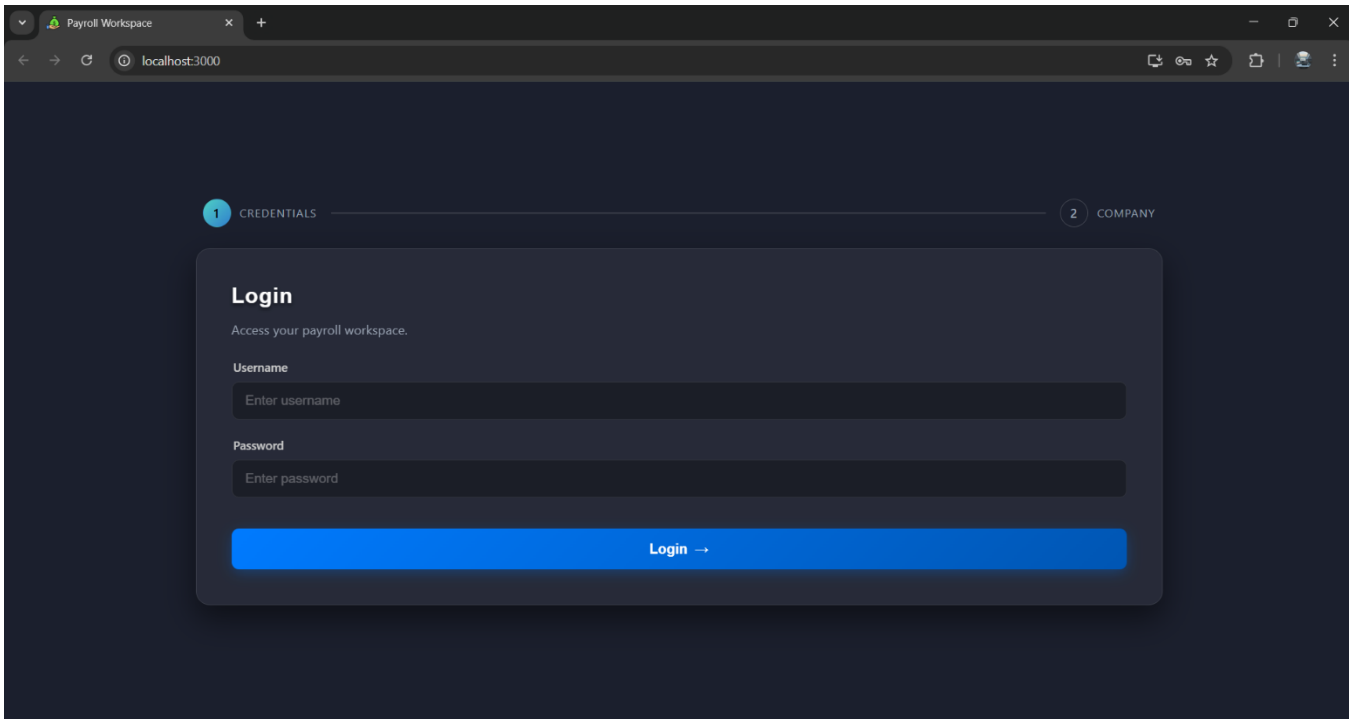
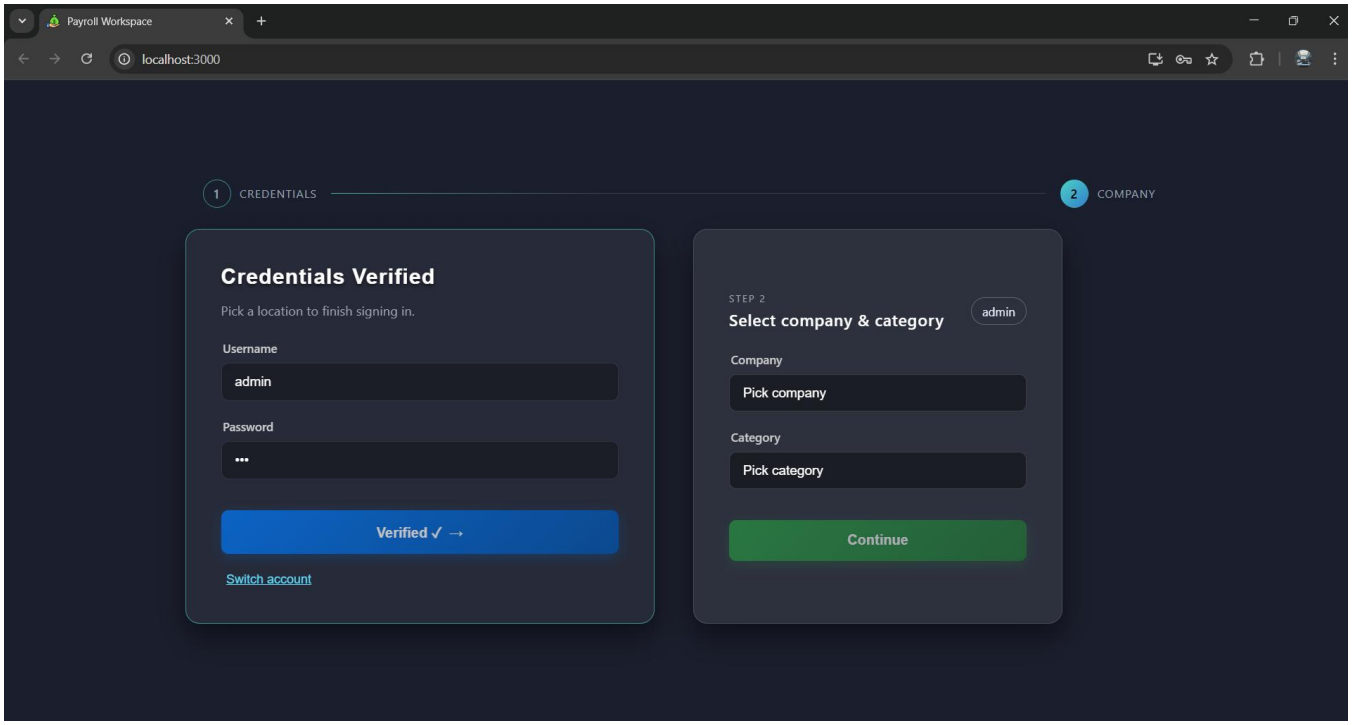
Web-Based Payroll System

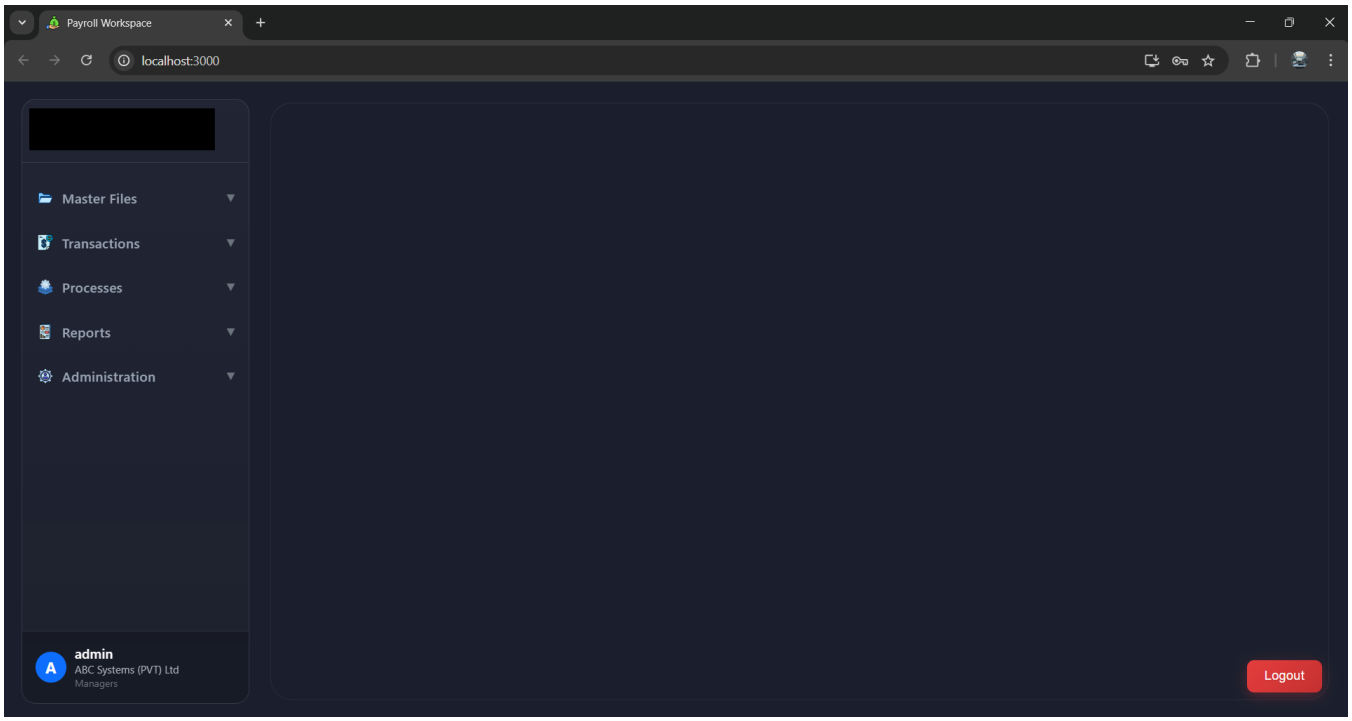
VERSION: 1.00

System Overview

The Web-Based Payroll System is a secure, responsive single-page application featuring an intuitive sidebar navigation and modal popups, allowing HR and administrative staff to use the system easily and effectively. The series of activities are categorized and arranged in an orderly, logical flow beginning with Master Files (for Company, Employee, and Bank configurations), followed by daily Transactions (such as loans and salary changes), proceeding to core Processes (for automated month-end calculations including EPF/ETF and dynamic PAYE), and concluding with Reports (digital payslips and bank transfer lists) respectively. This structured layout enables end-users to transition smoothly from legacy workflows and get to the desired screen easily and more quickly.




Payroll Workspace
localhost:3000
1 CREDENTIALS
2 COMPANY
Credentials Verified
Pick a location to finish signing in.
Username
admin
Password
•••
Verified ✓ →
Switch account
STEP 2
Select company & category
admin
Company
Pick company
Category
Pick category
Continue


Payroll Workspace
localhost:3000
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Master Files
Transactions
Processes
Reports
Administration
A
admin
ABC Systems (PVT) Ltd
Managers
Logout

Key System Features & Benefits

Automated Payroll Calculations - Efficiency and accuracy in payroll processing are critical. By automating the monthly calculation cycle, the system eliminates the need for manual, error-prone spreadsheet work. The platform seamlessly processes complex financial transactions, including basic pay adjustments, overtime, attendance allowances, and loan installments, saving the HR team significant time and administrative effort.

Dynamic Tax & Statutory Compliance - Computations for taxes and government statutory deductions can be complex and stressful. This system features automated tax management modules that minimize the burden of employee reporting obligations. All tax data and statutory deductions are calculated automatically and kept securely in the system, making it easy to pull accurate records for compliance without digging through piles of physical pay-slips.

Reliable Backup & Centralized Data Storing large amounts of sensitive payroll data manually or in fragmented files is risky. The application utilizes a centralized database to securely store all employee and financial records. With automated database backups, it is convenient to save records and ensures that, in the event of local hardware failure, all historical payroll data is preserved and fully recoverable.

Comprehensive & Instant Reporting - Generating required month-end documentation is reduced to a few clicks. The system automatically produces formatted digital payslips, customized bank transfer files, and mandatory government EPF/ETF reports. Historical data is securely archived upon month-end finalization, allowing HR to instantly pull past transaction records to quickly resolve any employee disputes.

Web-Based Accessibility & Usability - Unlike standalone desktop applications restricted to specific office computers, this modern web architecture allows authorized staff to securely manage payroll operations from anywhere using a standard web browser. The clean, responsive interface eliminates the steep learning curve of legacy software, making global employee updates and mass data management effortless.

Recommended Hardware & Software Requirements

Because this is a web-based application utilizing a Client-Server architecture, the system requirements are divided between the server environment (where the application and database are hosted) and the client environment (the end-users accessing the system).

Server Environment (Hosting)

Hardware Requirements:

- **Processor (CPU):** Quad-core processor (e.g., Intel Xeon or equivalent) @ 2.5 GHz or higher.
- **Memory (RAM):** 8 GB minimum (16 GB or higher recommended for optimal database performance during heavy month-end processing).
- **Storage:** 100 GB SSD (Solid State Drive) minimum for the operating system, application files, and database storage. SSD is highly recommended to ensure fast read/write speeds for database queries.
- **Network:** High-speed, stable internet connection with a static IP address.

Software Requirements:

- **Operating System:** Linux (e.g., Ubuntu Server 20.04 LTS or newer) or Windows Server (2019 or newer).
- **Web Server / Runtime:** Node.js environment (LTS version).
- **Database Management System:** MySQL (Version 8.0 or higher).
- **Security:** SSL/TLS Certificate for HTTPS encryption.

Client Environment (End-User)

Hardware Requirements:

- **Processor (CPU):** Dual-core processor (Intel Core i3 or equivalent).
- **Memory (RAM):** 4 GB minimum (8 GB recommended for smoother multitasking).
- **Display:** Monitor with a minimum resolution of 1366 x 768 pixels (1920 x 1080 Full HD recommended for optimal viewing of large data grids and reports).
- **Network:** Reliable internet or intranet connection to communicate with the application server.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or modern Linux distributions.
- **Web Browser:** An up-to-date modern web browser is required. Recommended browsers include:
  - Google Chrome (latest version)
  - Mozilla Firefox (latest version)
  - Microsoft Edge (latest version)
  - Apple Safari (latest version)
- **Additional Software:** A standard PDF reader (e.g., Adobe Acrobat or built-in browser PDF viewers) to view and print downloaded reports and payslips.

Non-Functional Requirements

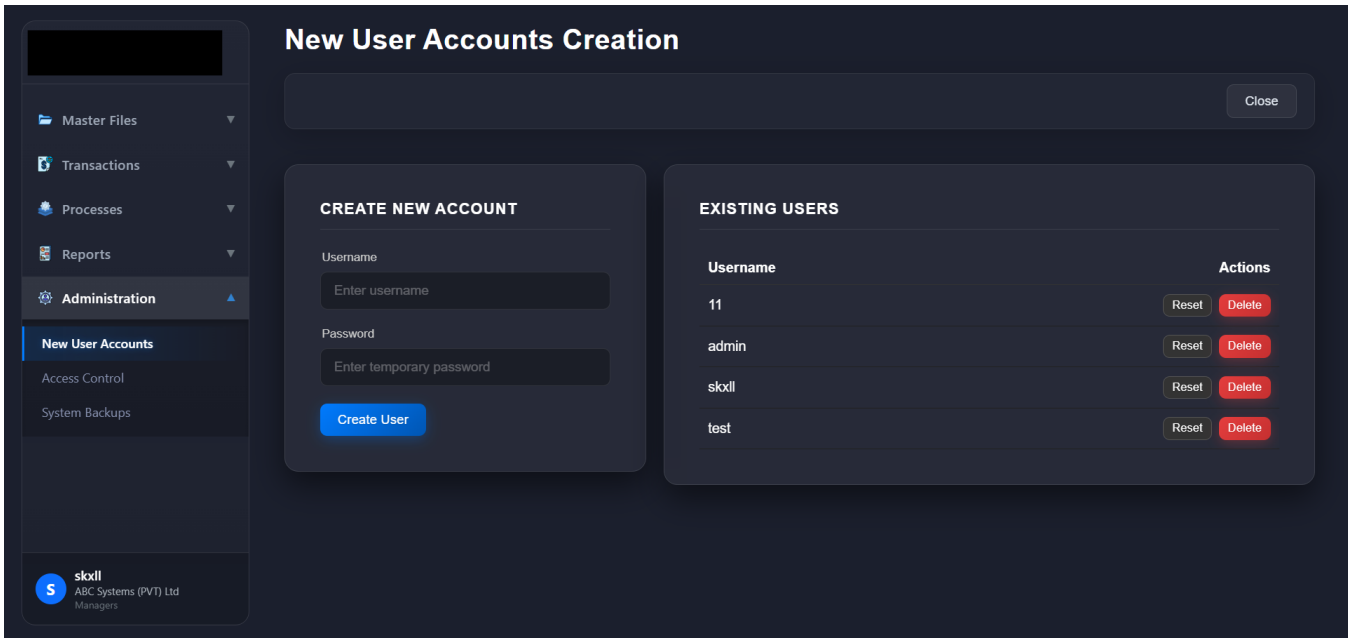
Security

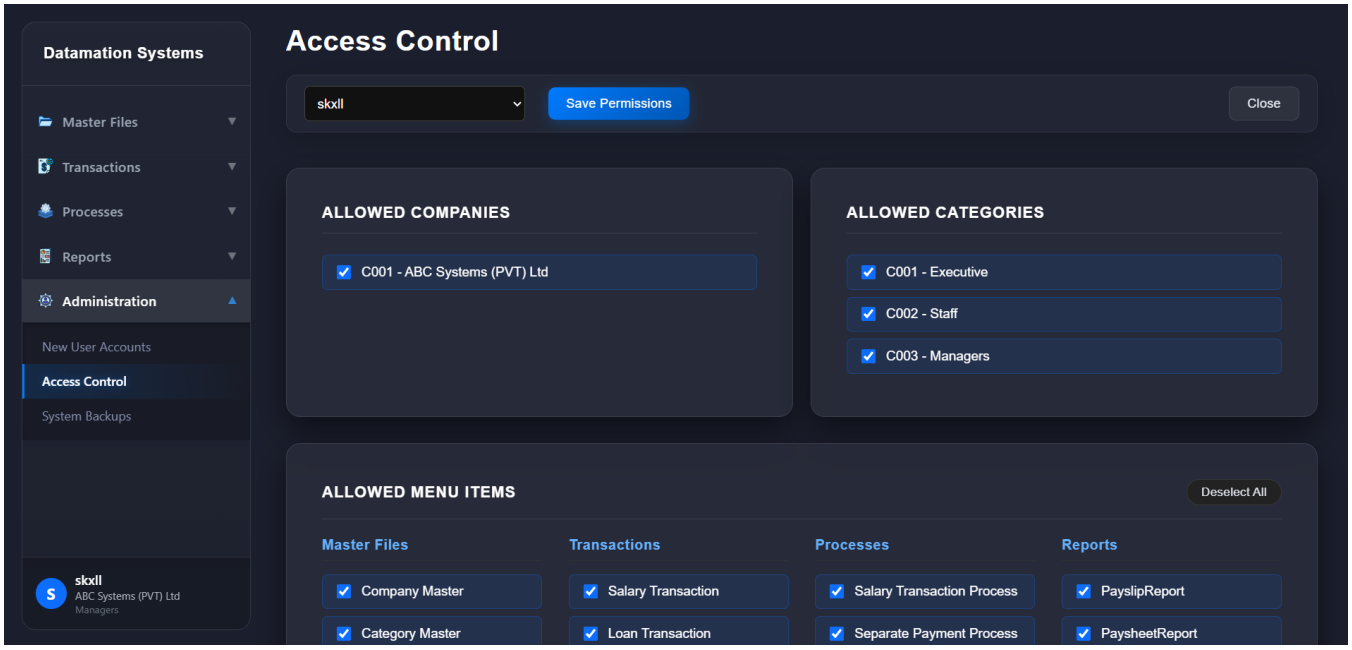
User Creation & Login to the System

- **Account Provisioning:** Only authorized System Administrators will have the capability to create, modify, or deactivate user accounts within the system.
- **Secure Authentication:** The login interface will require users to authenticate using uniquely assigned credentials (e.g., username/email and password). All user passwords must be securely encrypted and hashed before being stored in the database to prevent exposure in the event of a database breach.
- **Session Management:** The application will utilize secure token-based authentication to maintain user sessions. To prevent unauthorized access on unattended terminals, the system will automatically terminate the session and log the user out after a predefined period of inactivity.

Company Access Control

- **Role-Based Access Control (RBAC):** The system will enforce strict RBAC, allowing administrators to configure granular permissions. Users will only have access to the specific menus, screens, and functional operations (e.g., view-only vs. edit/delete capabilities) that are explicitly mapped to their assigned role.
- **Entity-Level Restrictions:** For organizations managing multiple branches, departments, or subsidiary companies, the application will feature dedicated 'Company Access Control'. This ensures that an HR representative or manager assigned to one specific company or department cannot view, edit, or process payroll data for another.
- **Data Segmentation:** Access privileges will be validated at the backend API level, ensuring that even if a user attempts to navigate to a restricted area directly, the system will block the request and protect the underlying data.


New User Accounts Creation
Close
Master Files
Transactions
Processes
Reports
Administration
New User Accounts
Access Control
System Backups
skxll
ABC Systems (PVT) Ltd
Managers
CREATE NEW ACCOUNT
Username
Enter username
Password
Enter temporary password
Create User
EXISTING USERS
Username
Actions
11
Reset
Delete
admin
Reset
Delete
skxll
Reset
Delete
test
Reset
Delete


Datamation Systems
Access Control
skxll
Save Permissions
Close
Master Files
Transactions
Processes
Reports
Administration
New User Accounts
Access Control
System Backups
skxll
ABC Systems (PVT) Ltd
Managers
ALLOWED COMPANIES
C001 - ABC Systems (PVT) Ltd
ALLOWED CATEGORIES
C001 - Executive
C002 - Staff
C003 - Managers
ALLOWED MENU ITEMS
Deselect All
Master Files
Transactions
Processes
Reports
Company Master
Salary Transaction
Salary Transaction Process
PayslipReport
Category Master
Loan Transaction
Separate Payment Process
PaysheetReport

Payroll System Operational Cycle

The steps to be followed to run payroll system for a particular month:

1. Check the system date in the Company master file.
2. Add new employees for this month and edit any existing employees' information in the employee master file.
3. Change any master file information if necessary.
4. Do employee promotion and transfers if available.
5. Enter all monthly data to the "salary transaction".
6. Print "Salary Transaction Listing" report & check for data entry errors.
7. Salary Increments
8. Amend or Add records to the loan transaction
9. Process "Salary Transaction Process"
10. Print all the monthly reports.
11. If user finds any data entry errors, correct it and run the month end again.
12. After finalizing the current month salary data print required reports.
13. Do the banks slip transfer process to send employee salary to bank.
14. EPF ETF e Transfer Process
15. After printing all the reports and to get prepare for the next month, run "Month End Process".